

TrES-1b

R-1118

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_201003 · created 2026-07-30T21:32:42+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459125.7439 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.048
Transit depth (Rp/Rs)^2	2.43 +/- 1.49 [%]
Orbital Inclination (inc)	89.74 +/- 1.38
Airmass coefficient 1 (a1)	1.632 +/- 0.034
Airmass coefficient 2 (a2)	-0.1595 +/- 0.0071
Scatter in the residuals of the lightcurve fit is	1.95 %
Best Comparison Star	None
Optimal Aperture	4.29
Optimal Annulus	12.47
Transit Duration (day)	0.0811 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.156 $\pm$ 0.048 vs 0.1358 (deviation 0.42 σ)
Duration fit vs published	0.0811 d vs 0.1042 d (deviation 6.07 σ)
Mid-time O-C	-0.2 min
Depth SNR	3.2
Residual scatter	1.95 %

Light curve

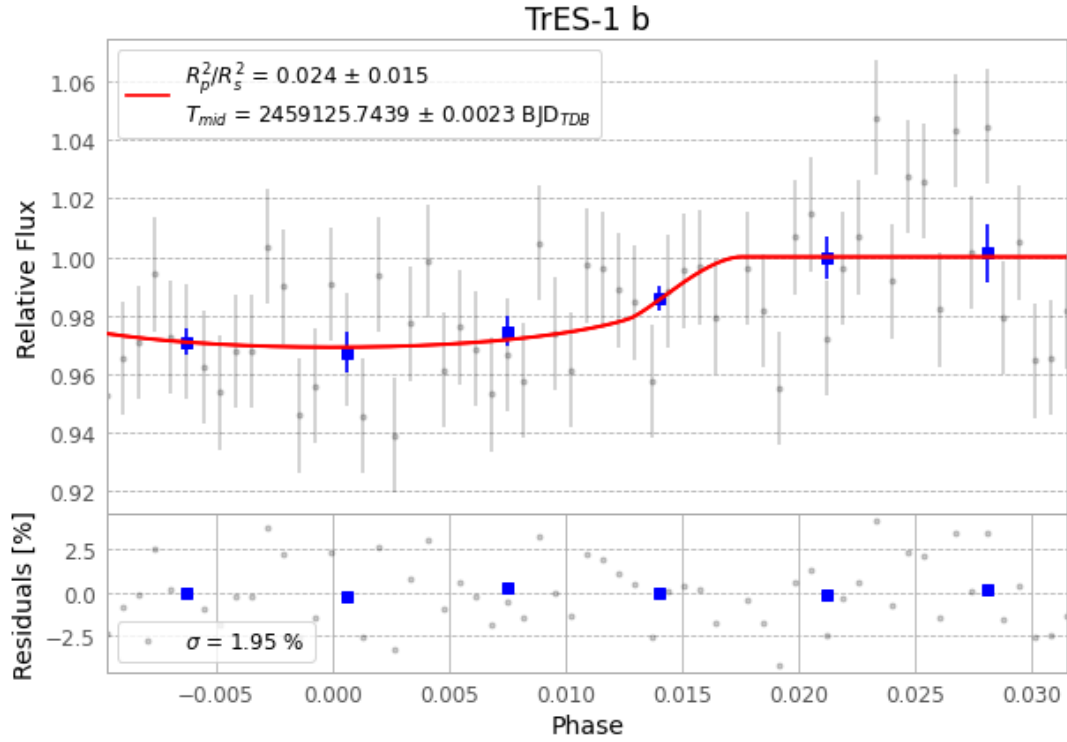


Figure 1 — FinalLightCurve_TrES-1 b_2020-10-02.png (SHA-256 7cb8995664dfa689...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [253, 235], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 053a4ce32b0897c1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

61 FITS frames (40 MB), TRES-1201003050612.FITS → TRES-1201003080613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-201003.log (eae56c248d13...), FinalLightCurve_TrES-1 b_2020-10-02.png (7cb8995664df...), FinalLightCurve_TrES-1 b_2020-10-02.csv (5286692d65c3...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 21:32Z.